

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

ROTATIONAL MOTION

LECTURE-05 PHYSICS

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Topics To Be Covered

- 1 Revision
 - + Radium of Gyration
 - + Comparison Translation and Rotation
- 2 Law of Conservation of Momentum
- 3 Centre of Gravity
- 4 Numericals
- 5 ..

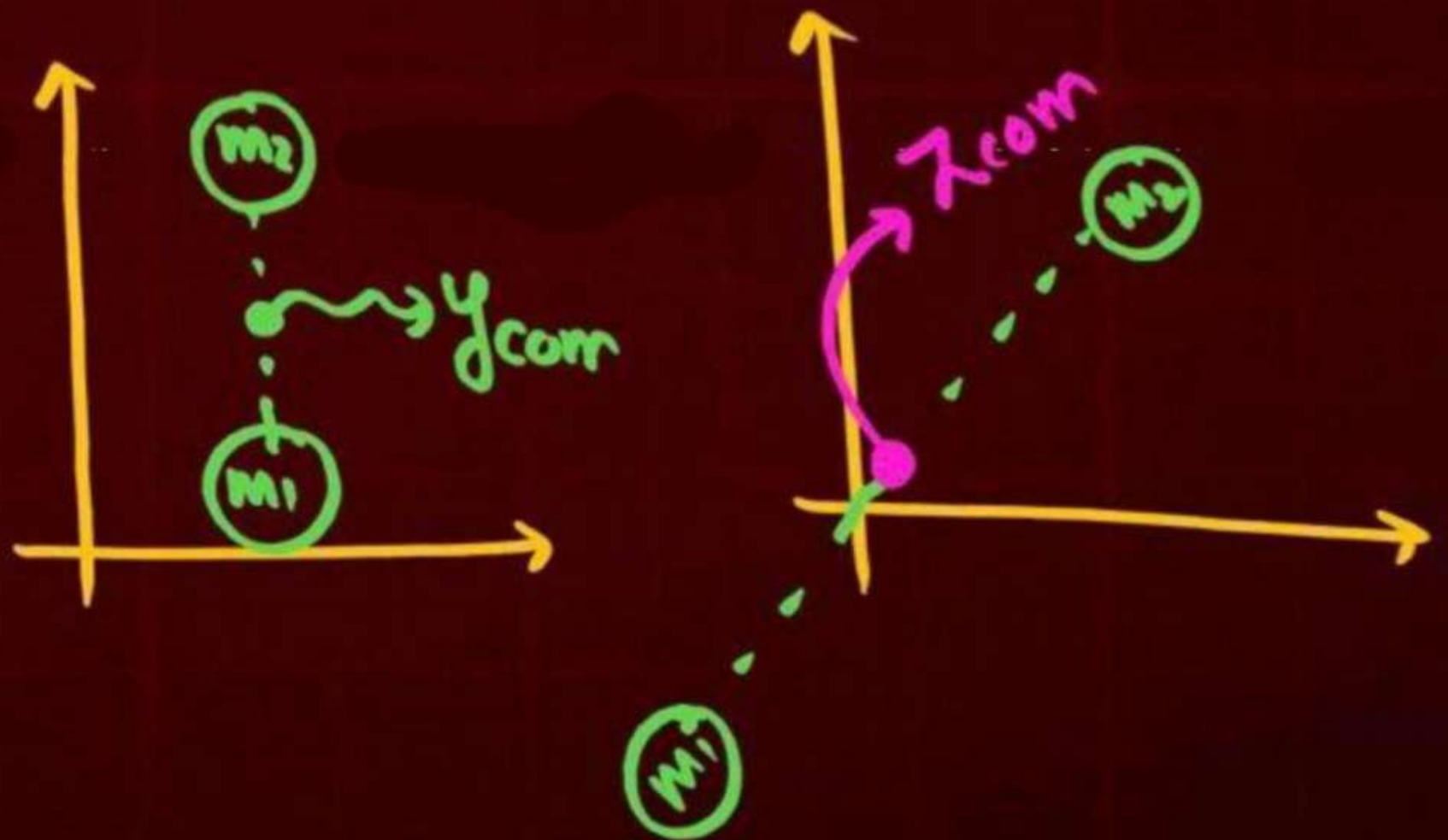
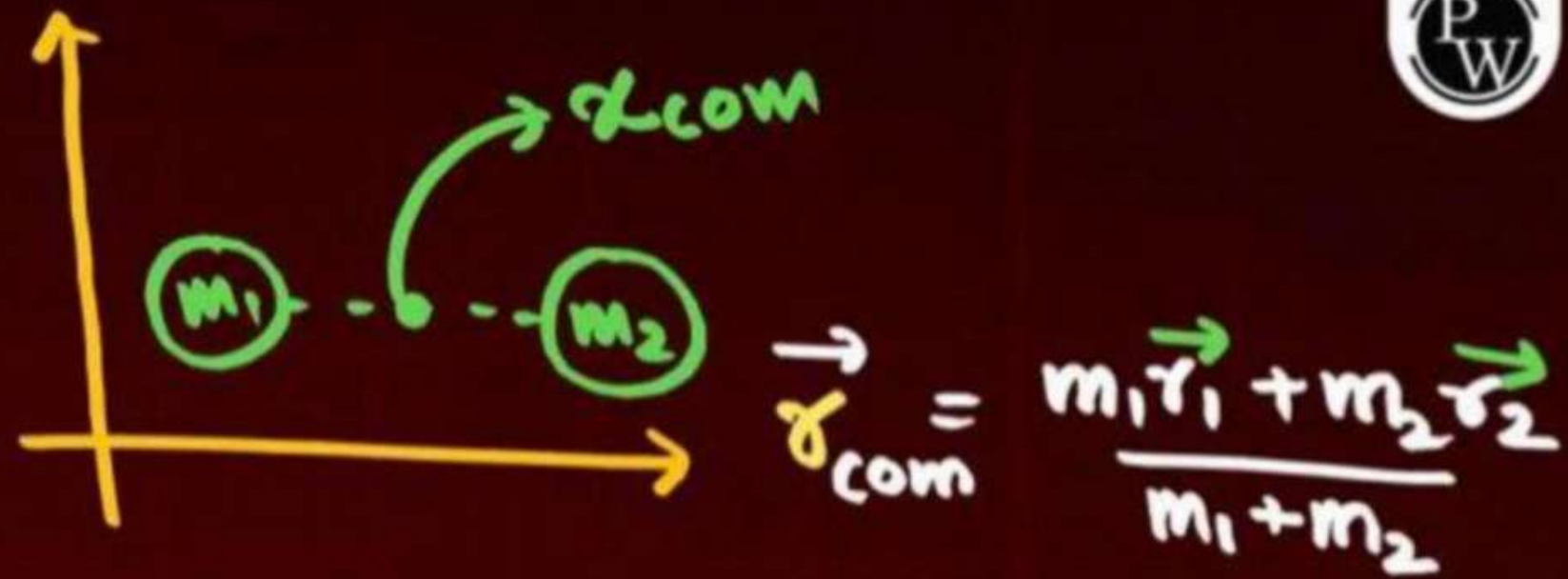
Recap From 1st Lecture

Position of Com

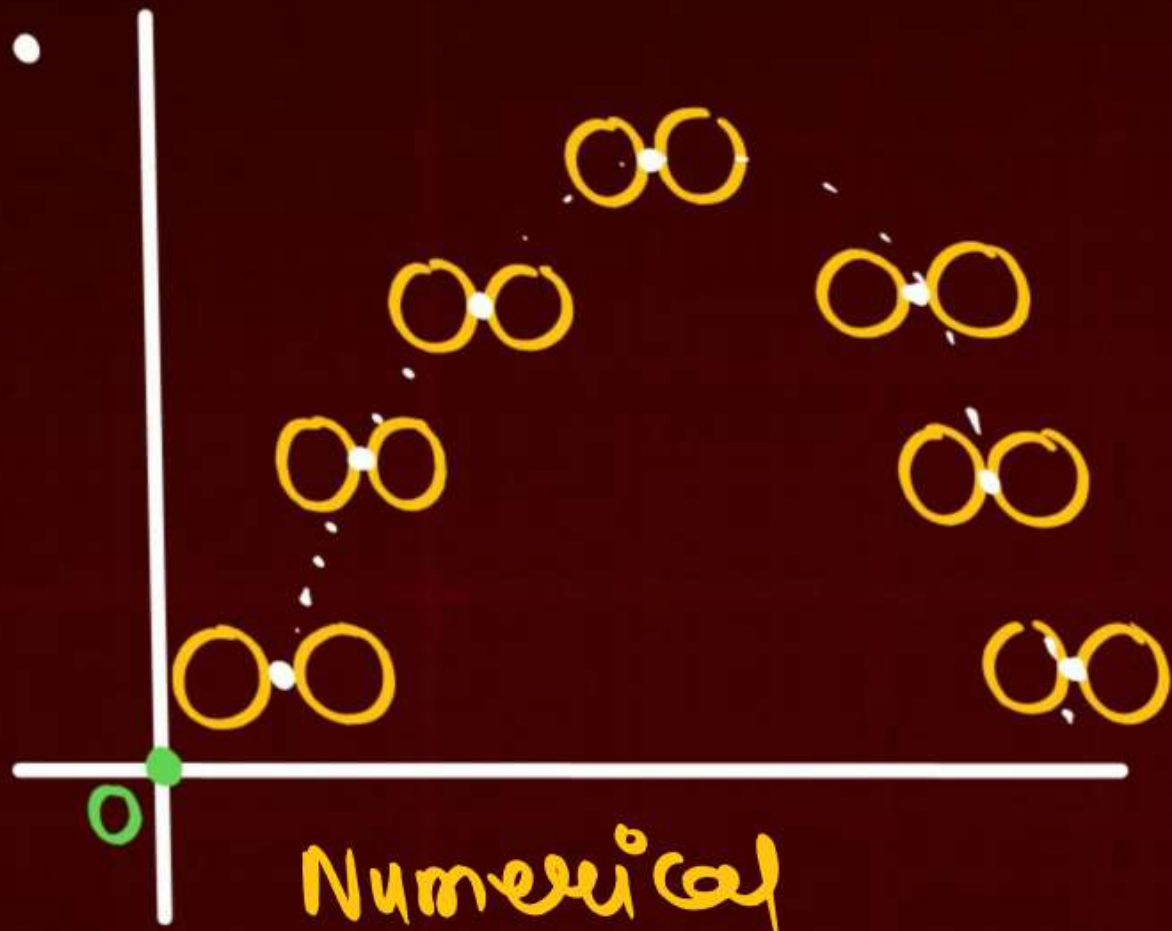
$$x_{\text{com}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$y_{\text{com}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$z_{\text{com}} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$



RECAP FROM 2nd LECTURE

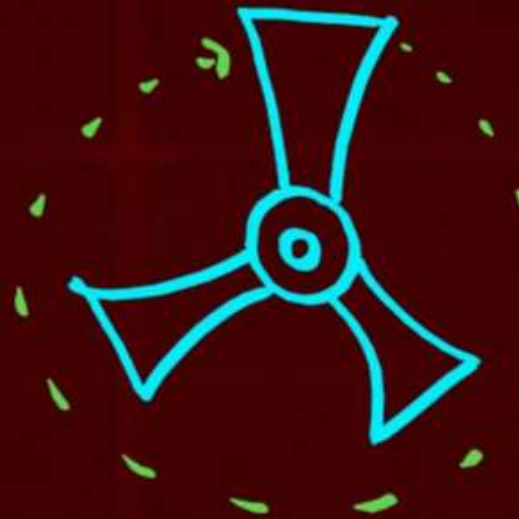


Numerical

$$\textcircled{1} V_{com} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$\textcircled{2} a_{com} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$$

Rotational Motion



✓ Angular displacement

$$\theta = \frac{\text{Arc}}{R} \quad x = R\theta \quad \text{Rad}$$

✓ Angular velocity

$$\omega = \frac{\theta}{t} \quad v = R\omega \quad \text{Rad/sec}$$

✓ Angular acceleration

$$\alpha = \frac{\omega}{t} \quad a = R\alpha \quad \text{Rad/sec}^2$$

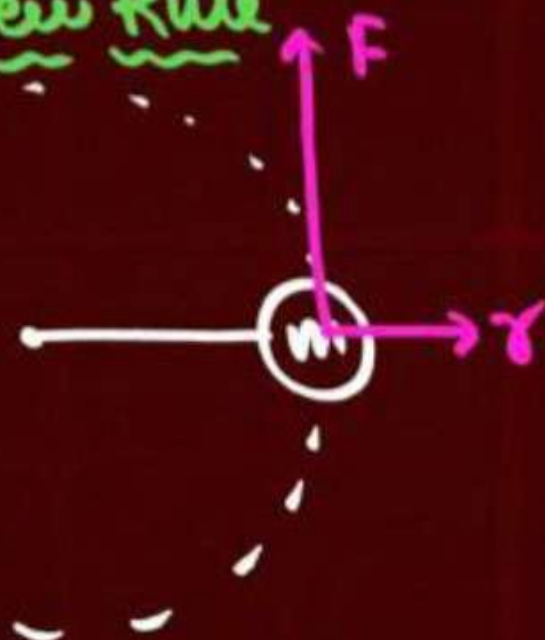
Recap From 3rd Lecture

Angular momentum

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = r p \sin \theta \hat{n}$$

Screw Rule



Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = r F \sin \theta \hat{n}$$

numerical

$$\checkmark \vec{r} = 2\hat{i} + 3\hat{j} \quad \vec{p} = 3\hat{i} + 4\hat{j}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\checkmark \vec{r}' = 2\hat{i} + 3\hat{j} + 5\hat{k} \quad \vec{p}' = 3\hat{i} + 4\hat{j} + 3\hat{k}$$

$$\vec{L}' = \vec{r}' \times \vec{p}'$$

Trick

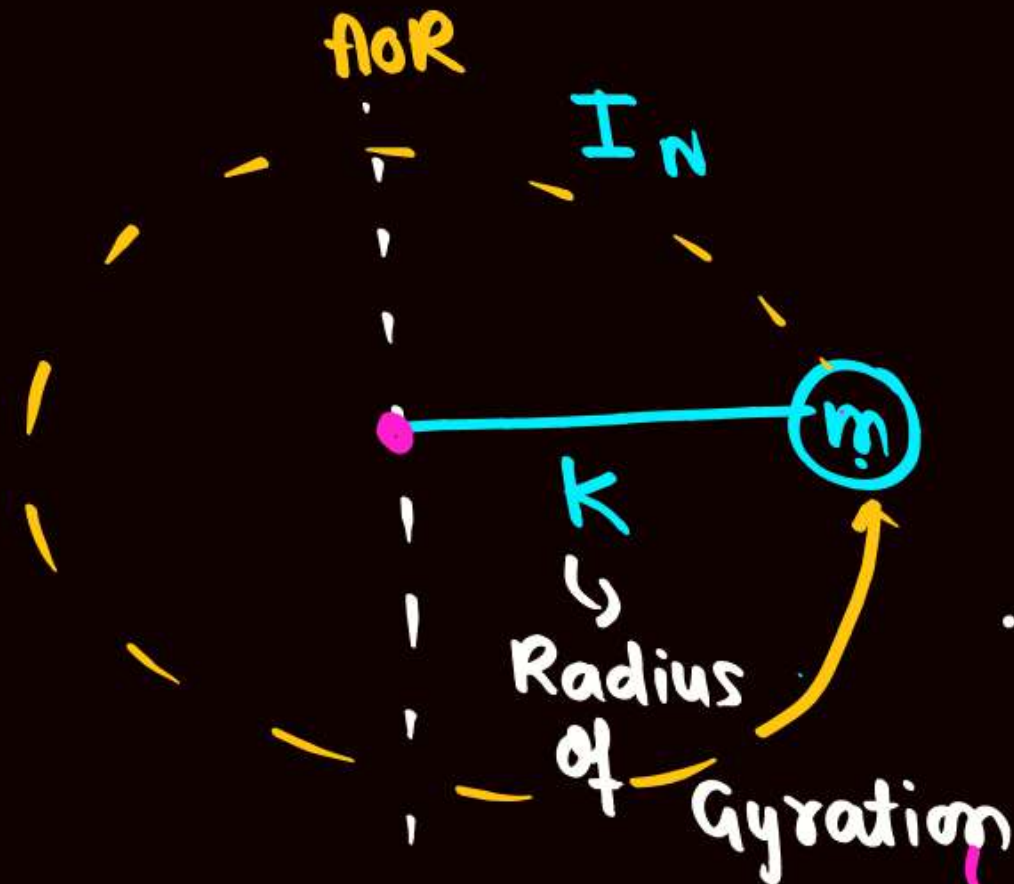
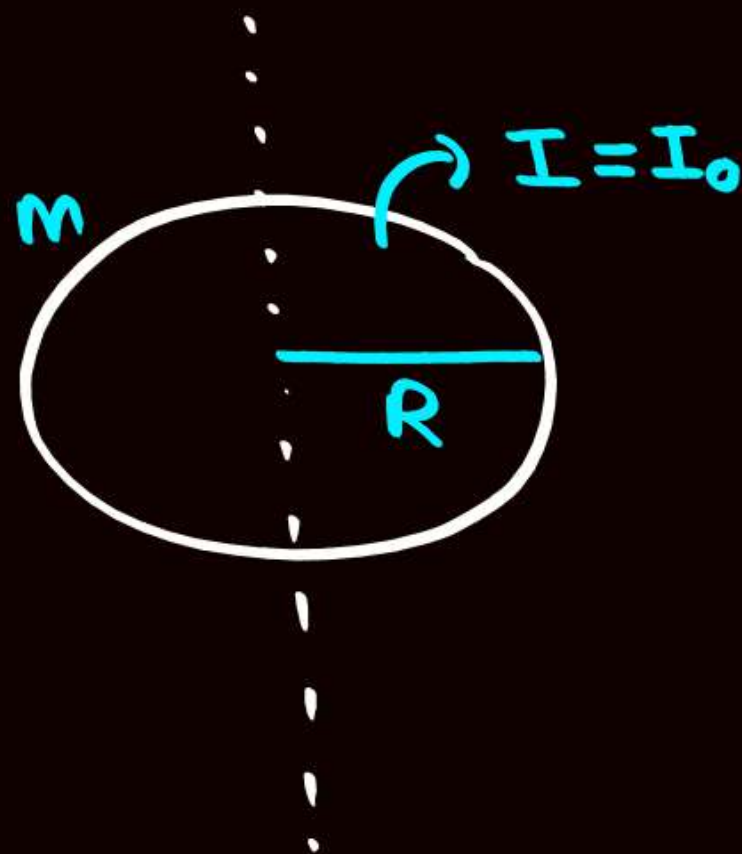
lecture no ⑤

Radius of Gyration

✓ Imp ~ JKBOSE (2024, 2023, 2021, 2019) - ③M

$$I = mR^2$$
$$I_N = mk^2$$

② The Radius of gyration is a distance from Axis of Rotation to the point where whole mass is concentrated without changing its momentum inertia



$$I_0 = I_N$$

$$I = mk^2$$

$$\frac{I}{m} = k^2$$

$$\sqrt{\frac{I}{m}} = k$$

Comparison b/w Translation and Rotational Motion

Translation



① $x \Rightarrow$ displacement

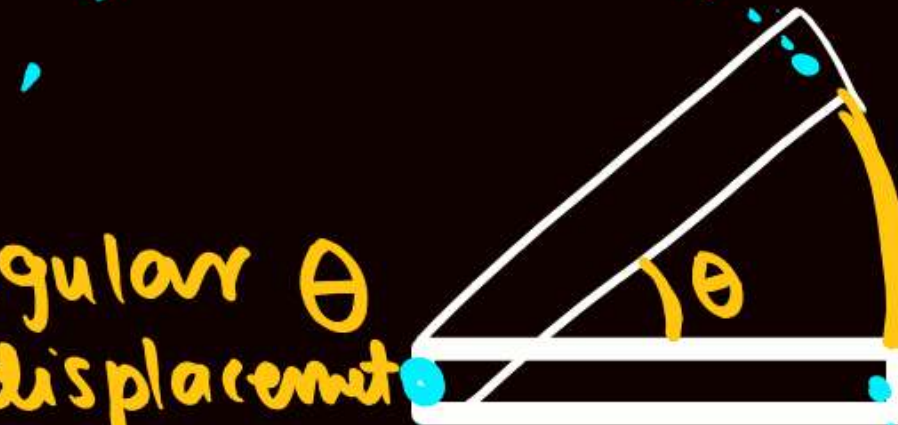
② time $\rightarrow t$

③ Velocity $v = \frac{x}{t}$

④ accelerating $\vec{a} = \frac{\Delta v}{t}$

⑤ mass (m)

Rotational



① Angular displacement θ

② time $\rightarrow t$

③ Angular velocity $\omega = \frac{\theta}{t}$

④ Angular acc $\alpha = \frac{\omega}{t}$

⑤ moment of

(I) Inertia = $m R^2$

ω_0
Initial Angular velocity

ω
Final Angular velocity

Comparison blw Translation and Rotational Motion

Translation

6. Force $\underline{F = ma}$

④ momentum $\vec{p} = m\vec{v}$

⑧ work = $f \times$

mca

⑨ K.E = $\frac{1}{2}mv^2$

mca
2023

Rotational

6. Torque $\tau = \vec{r} \times \vec{F}$, $\tau = \frac{d\vec{L}}{dt}$

$\tau = I\alpha$ ✓

⑦ angular momentum

$\vec{L} = \vec{r} \times \vec{p}$, $\vec{L} = I\omega$

⑧ ✓ $\omega = \tau \theta$

⑨ K.E = $\frac{1}{2}I\omega^2$
Rotation

T-3 Kinematics Equation for Rotational Motion

we have three equation of motion (1D motion)



Translation

$$\textcircled{1} \underline{v} = \underline{u} + \underline{at}$$

$$\textcircled{2} \overset{x}{\underline{s}} = \underline{u}t + \frac{1}{2}at^2 \quad \textcircled{3} \underline{v}^2 = u^2 + 2a\underline{s} \quad \rightarrow x$$

Rotation

$$\textcircled{1} \omega = \omega_0 + \alpha t$$

1st equation of motion in Rotation

$$\textcircled{2} \theta = \omega_0 t + \frac{1}{2}\alpha t^2 \quad \textcircled{3} \omega^2 = \omega_0^2 + 2\alpha\theta$$

2nd equation

3rd equation

Law of conservation of Angular momentum

when external Torque is Zero

“Then initial angular momentum is equal to the Final Angular momentum”

Chota
change

Maths
we know that

$$\tau = \frac{dL}{dt}$$

$$0 = \frac{dL}{dt}$$

$$0 \times dt = dL$$

$$0 = dL$$

$$\rightarrow dL = 0$$

$$\Delta L = 0$$

$$\Delta L = 0$$

$$L_f - L_i = 0$$

$$\boxed{L_f = L_i}$$

$$L_i = L_f$$

$$L_i = L_f \quad (\tau_{\text{ext}} = 0)$$

$$p = m \underline{v}$$

$$p_i = m_i v_i$$

$$p_f = m_f v_f$$

$$L = I \omega$$

$$L_i = I_i \omega_i$$

$$L_f = I_f \omega_f$$

$$I_i \omega_i = I_f \omega_f$$

$$\uparrow \downarrow \textcircled{I_i} = \frac{I_f \omega_f}{\downarrow \textcircled{\omega_i} \uparrow}$$

$$\downarrow \uparrow I \propto \frac{1}{\downarrow \omega \uparrow}$$

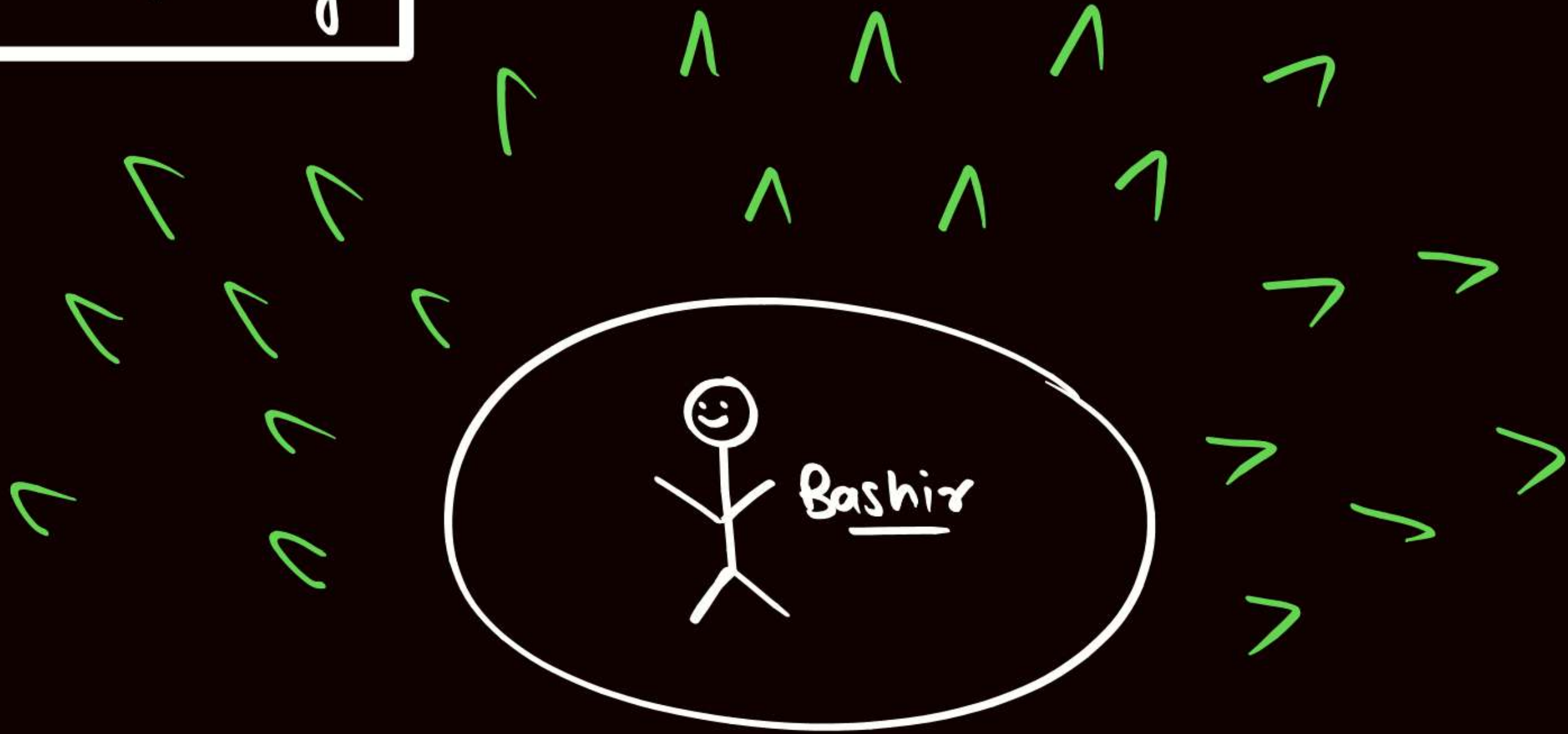
$$I \uparrow \quad \omega \downarrow$$

$$I \downarrow \quad \omega \uparrow$$

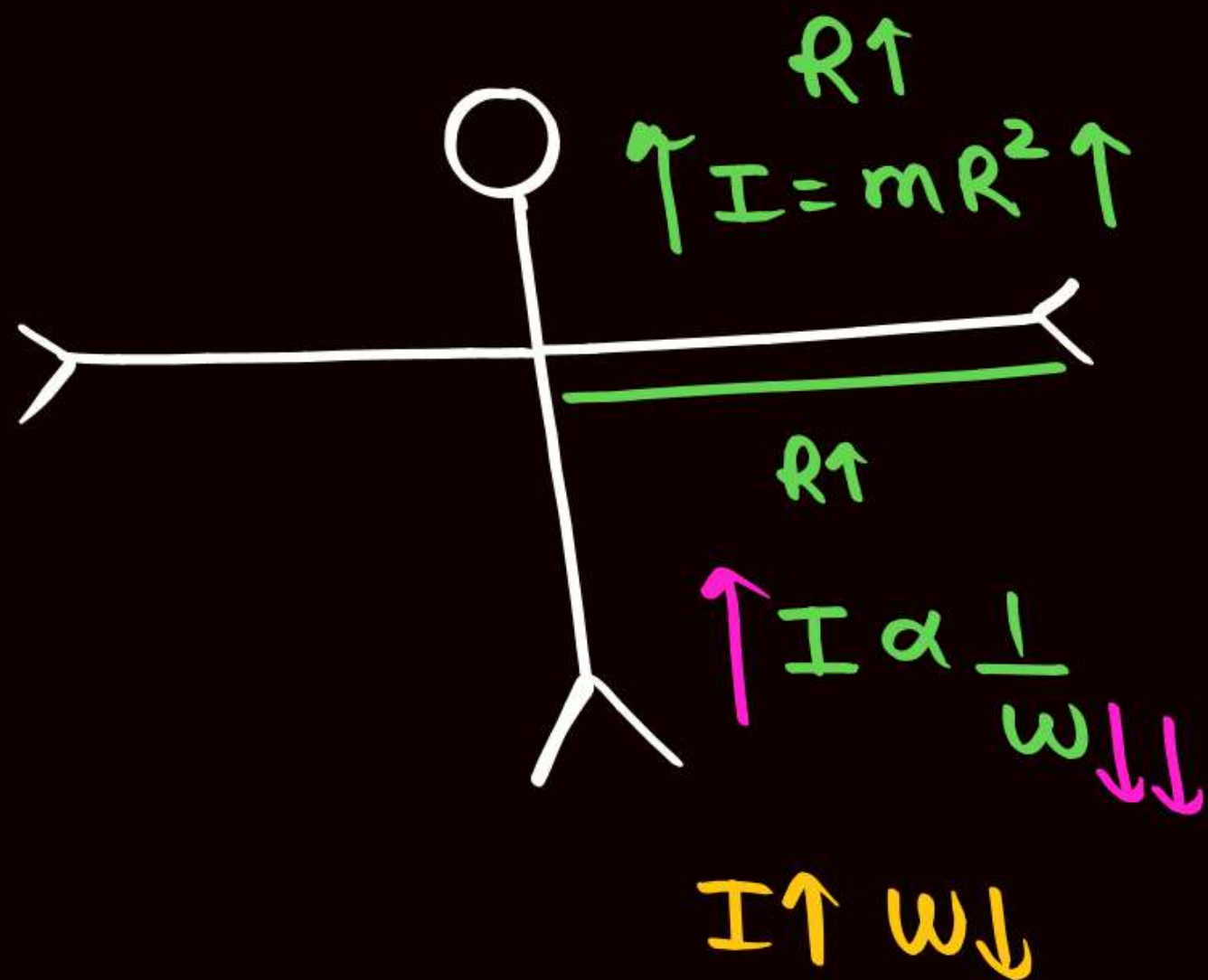
$\Rightarrow$ Application
of
Law of conservation
of Angular momentum

(Application of law of conservation of $\vec{L}$)

(i) ice skating



Part - I



Part II



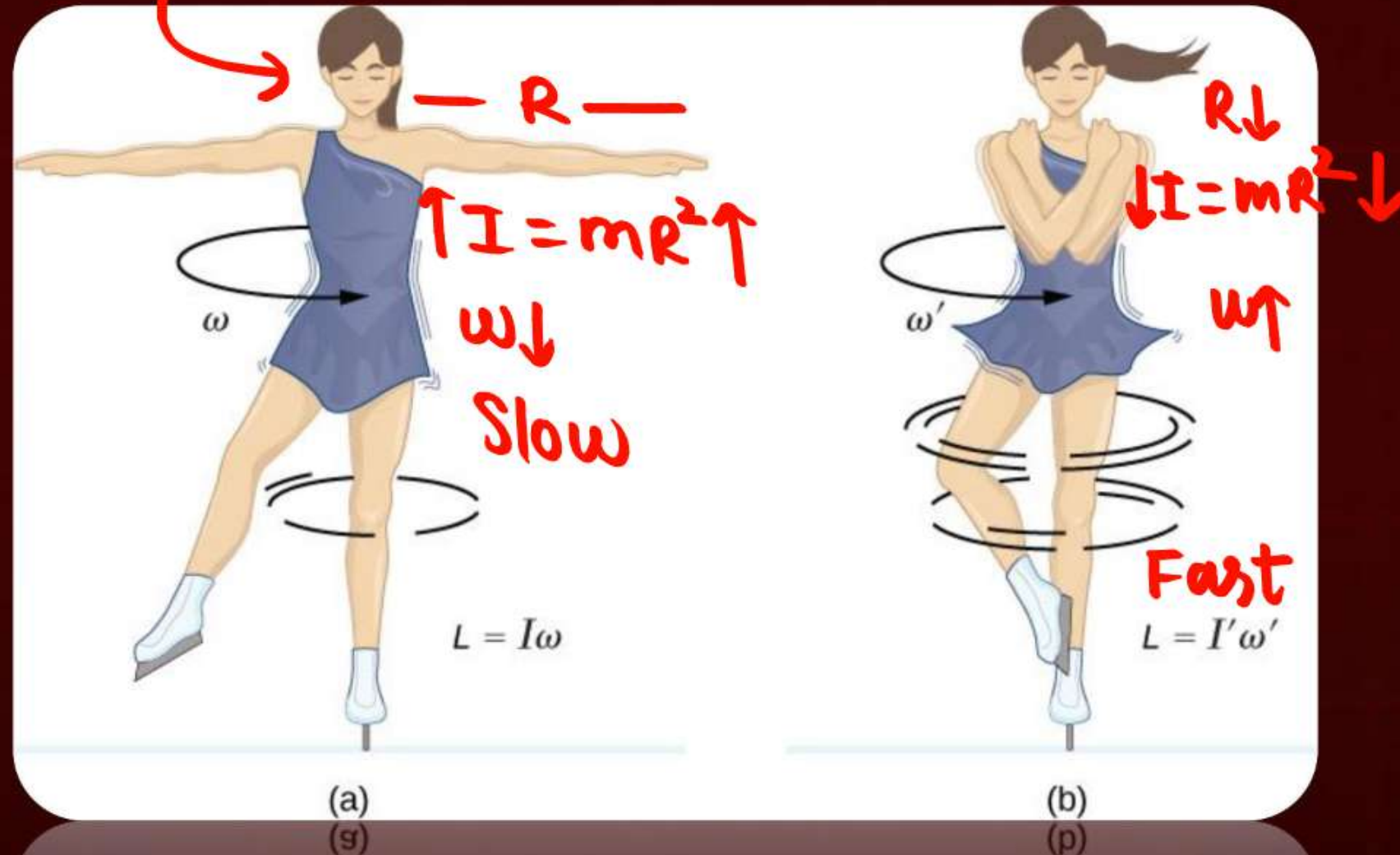
Law of Conservation of Momentum

When external torque is zero then initial angular momentum is equal to final angular momentum

L_i

L_f

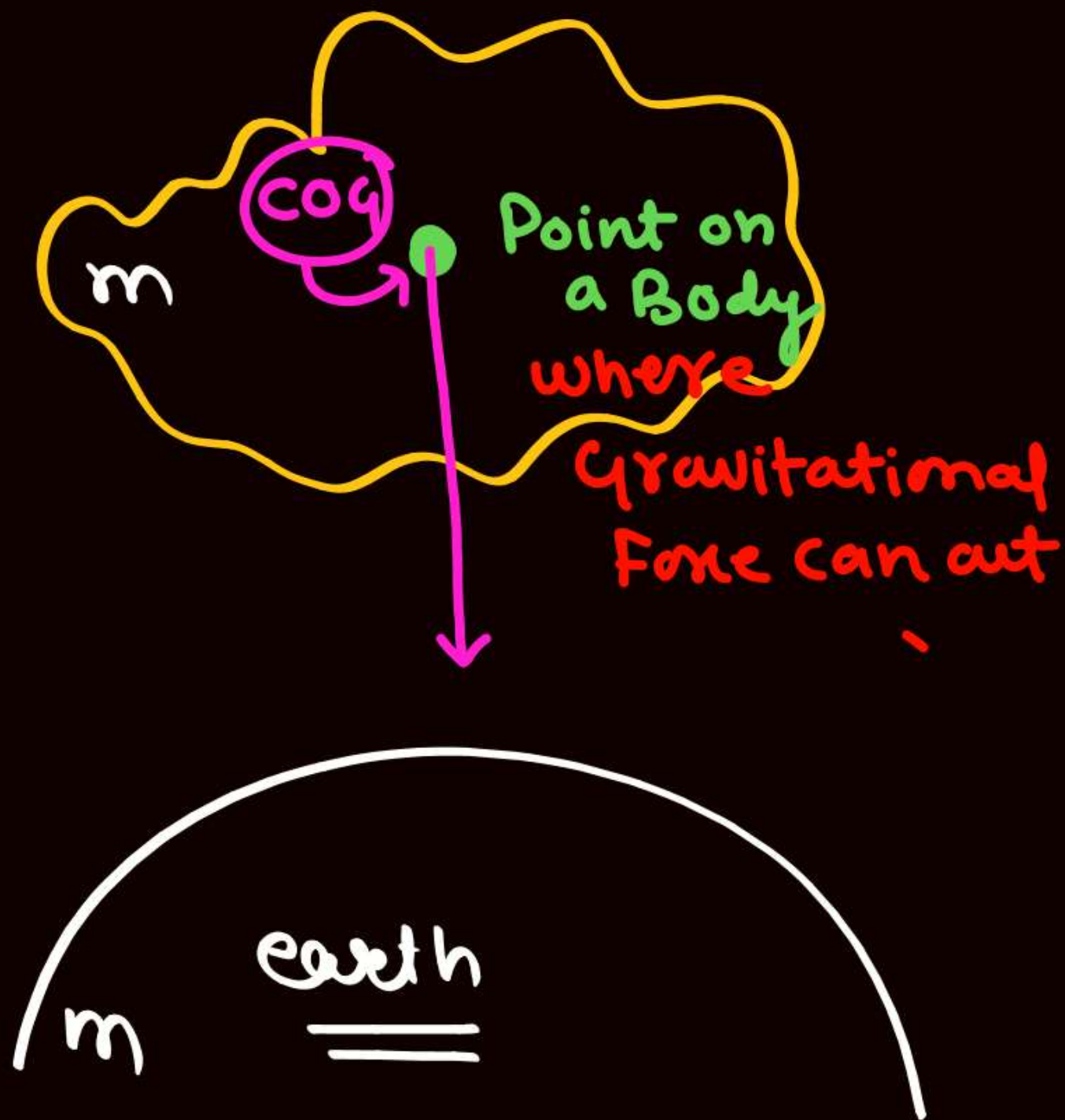
Bashir



2014

Centre of Gravity

The Centre of Gravity of a Body is that point where gravitational force act on it



Numerical.

Q If the initial moment of inertia (I_i) and initial angular velocity are $\underline{2 \text{ Kg m}^2}$ and $\underline{4 \text{ rad/s}}$ **no external Torque is Acting** calculate $\textcircled{I_f}$ If its Final angular velocity is $\underline{8 \text{ rad/s}}$? $\underline{JKCET = 2015}$

(a) 1 Kg m^2

(b) 2 Kg m^2

(c) 3 Kg m^2

(d) 4 Kg m^2

Sol

ω_f

$$\underline{\underline{\text{Sol}}}$$
$$L_i = L_f$$

$$\underline{I_i \omega_i} = \underline{I_f \omega_f}$$

$$\underline{2 \times 4} = I_f \times 8$$

$$8 = I_f \times 8$$

$$\frac{8}{8} = I_f$$

$$1 \text{ kg m}^2 = I_f$$

Kar Ke Dikhao

H/W



What is meant by rotational motion?

Ye asan hai?

What is meant by moment of mass of a particle?

Can you solve this?

Can centre of mass be outside the body?

Bolo kise pata hai?

Write the formula of position vector of centre of mass of two particles.

Chlo Start krte hai

Write the position of centre of mass of a uniform thin rod.

Chlo inka relation dekhte hai

Write the relation between τ and L .

Gravity ye kya hoti hai?

What is meant by centre of gravity?

Chlo smjhte hai

State the dependence of moment of inertia. $I = mR^2$

Chaliye Shuru krte hai

Find MOI of a disc of radius 0.8 m and mass is 2 kg. When AOR is passed through perpendicular from center of disc. ✓

Solution:

$$M = 2 \text{ kg}$$

$$r = 0.8$$

$$I = ?$$

$$I = \frac{MR^2}{2}$$

Ye Easy hai?

Find MOI of a disc of radius 0.8 m and mass is 2 kg. When AOR is passed through perpendicular from center of disc.

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Thank
You

